1 **(a)** 125 **[B1]**

(b) Corresponding angles (are equal) **[B1]**

(c) 144 **[B1]**

[Total Marks: 3]

2 **(a)** Identifies alternate angles by drawing a parallel line through E OR forms a triangle by extending a line **[M1]**

85 **[A1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

[Total Marks: 4]

3 **(a)** $360 \div 24$ **[M1]**

15 **[A1]**

(b) $(15 - 2) \times 180$ **[M1]**

2340 **[A1]**

(c) Exterior angle $= 180 - 165 = 15^\circ$ **[M1]**

$360 \div 15 = 24$ **[A1]**

[Total Marks: 6]

4 **(a)** $180 - (55 + 70)$ **[M1]**

55 **[A1]**

(b) 55 **[B1]**

(c) Angle $DAE = 55^\circ$ (from part a) AND Angle $ADE = 55^\circ$ **[B1]**

Two angles are equal, so the triangle is isosceles **[B1]**

[Total Marks: 5]

5 **(a)** Equates sum of angles to formula: $3 \times 140 + 165(n - 3) = 180(n - 2)$ **[M1]**

Expands brackets correctly: $420 + 165n - 495 = 180n - 360$ **[M1]**

Rearranges to isolate n: $15n = 285$ (or equivalent) **[M1]**

19 **[A1]**

(b) States $a + b + c = 180$ **[B1]**

Substitutes $a + b = c$ to get $c + c = 180 \implies 2c = 180 \implies c = 90$ completing the proof **[B1]**

(c) **(i)** Calculates exterior angle of $P = \dfrac{360}{12} = 30^\circ$ so exterior angle of $Q = 15^\circ$ **[M1]**

24 **[A1]**

(ii) Calculates interior angles of P and Q: $180 - 30 = 150^\circ$ and $180 - 15 = 165^\circ$ **[M1]**

$360 - (150 + 165) = 45$ **[A1]**

[Total Marks: 10]

6 **(a)** Recognises the sum of the interior angles of a pentagon is 540° **[B1]**

Sets up a correct equation: $(2x + 10) + 100 + 80 + 3x + (x + 50) = 540$ **[M1]**

50 **[A1]**

(b) Sets up a relationship for the angles, e.g., $4E + E = 180$ where E is the exterior angle **[M1]**

$E = 36^\circ$ **[M1]**

10 **[A1]**

[Total Marks: 6]

7 **(a)** $5x + 10 = 7x - 30$ **[M1]**

$x = 20$ **[A1]**

Corresponding angles (are equal) **[B1]**

(b) Angle $GHD = 7(20) - 30 = 110^\circ$ **[B1]**

Angle $GHK = 180 - 110 = 70^\circ$ (Angles on a straight line) **[B1]**

Angle $GKH = 180 - (70 + 35) = 75^\circ$ (Angles in a triangle) **[B1]**

[Total Marks: 6]

8 **(a)** $\dfrac{(5-2) \times 180}{5}$ OR $180 - \dfrac{360}{5}$ **[M1]**

108 **[A1]**

(b) Interior angle of hexagon $= 120^\circ$ **[B1]**

Angle $AEI = 360 - 108 - 120 = 132$ **[A1]**

(c) $AE = EI$ (sides of regular polygons sharing an edge are equal) **[M1]**

Triangle AEI is isosceles **[M1]**

Angle $EAI = \dfrac{180 - 132}{2} = 24$ **[A1]**

[Total Marks: 7]

9 **(a)** $(5 - 2) \times 180 \div 5$ OR $180 - (360 \div 5)$ **[M1]**

108 **[A1]**

(b) Interior angle of square $= 90^\circ$ **[B1]**

Interior angle of n-gon $= 360 - 90 - 108 = 162°$ **[M1]**

$n = 360 \div (180 - 162) = 20$ **[A1]**

(c) Recognises triangle PFA is isosceles (since $PA = PF$ from shared sides of regular polygons) **[M1]**

Angle $PFA = (180 - 162) \div 2 = 9$ **[A1]**

[Total Marks: 7]

10 (a) 48 **[B1]**

(b) 56 **[B1]**

(c) $180 - 48 - 56 = 76$ **[B1]**

[Total Marks: 3]

11 (a) Sum of interior angles of a pentagon $= (5 - 2) \times 180° = 540°$ **[M1]**

$x + (x + 20) + (2x - 10) + (2x + 30) + 140 = 540$ **[M1]**

$6x + 180 = 540 \implies 6x = 360 \implies x = 60$ **[A1]**

(b) Substituting $x = 60$ into $(2x + 30)$ or checking all angle sizes $(60°, 80°, 110°, 150°, 140°)$ **[M1]**

150 **[A1]**

[Total Marks: 5]

12 (a) Recognises that $\angle PXQ = \angle BPX + \angle DQX$ (e.g. by drawing a parallel line through X to form alternate angles) **[M1]**

$(3x + 15) + (2x + 10) = 75$ **[M1]**

$5x + 25 = 75 \implies 5x = 50 \implies x = 10$ **[A1]**

(b) Expression for exterior or interior angle of P: $\dfrac{360}{n}$ or $180 - \dfrac{360}{n}$ **[B1]**

Expression for exterior or interior angle of Q: $\dfrac{360}{3n}$ or $180 - \dfrac{120}{n}$ **[B1]**

Forms a correct equation: $\left(180 - \dfrac{120}{n}\right) - \left(180 - \dfrac{360}{n}\right) = 16$ OR $\dfrac{360}{n} - \dfrac{120}{n} = 16$ **[M1]**

$\dfrac{240}{n} = 16 \implies n = 15$ **[A1]**

[Total Marks: 7]